

FedASA: A Personalized Federated Learning With Adaptive Model Aggregation for Heterogeneous Mobile Edge Computing

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in
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CERTIFICATE

This is to certify that the report entitled **FedASA: A Personalized Federated Learning With Adaptive Model Aggregation for Heterogeneous Mobile Edge Computing** submitted by **NOBIN SIJO** (KNP23CS082), to the Department of Computer Science and Engineering in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in **Computer Science and Engineering** of the APJ Abdul Kalam Technological University is a bonafide record of the seminar work carried out by him under our guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

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I, **NOBIN SIJO** (KNP23CS082), hereby declare that the seminar report entitled **FedASA: A Personalized Federated Learning With Adaptive Model Aggregation for Heterogeneous Mobile Edge Computing**, submitted for partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering of the APJ Abdul Kalam Technological University, Kerala, is a bonafide work done by me under the supervision of **APARNA M** (Assistant Professor, Dept. of CSE, College of Engineering Karunagappally).

This submission represents my ideas in my own words, and where ideas or words of others have been included, I have adequately and accurately cited and referenced the original sources.

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Abstract

Federated Learning (FL) is an indispensable decentralized paradigm for the Industrial Internet of Things (IIoT) and Mobile Edge Computing (MEC), enabling privacy-preserving collaborative model training across distributed edge nodes. However, deploying conventional FL in industrial edge environments is hindered by a crippling trilemma: **statistical heterogeneity**, **device resource constraints**, and **client unfairness**. Existing methods focus primarily on cross-domain variations (operational conditions such as varying motor speeds and loads) while overlooking **cross-location heterogeneity** (discrepancies arising from differing physical sensor mounting coordinates), which causes catastrophic client drift and severe model degradation. Furthermore, monolithic all-sharing models impose immense communication overhead on bandwidth-limited edge channels, while fixed layer-sharing (*G*-sharing) architectures fail to handle diverse edge distributions.

To address these fundamental challenges, this report investigates **FedASA**, a personalized federated learning framework featuring adaptive model aggregation. FedASA introduces a *cell-wise shared architecture selection mechanism* that dynamically determines the optimal partition between shared and personalized parameters for each edge device using an auxiliary alignment criterion under communication constraints. Additionally, an *operator-level cell-based aggregation algorithm* enables parameter servers to aggregate heterogeneous modular network architectures, generalizing canonical Federated Averaging (FedAvg). Rigorous theoretical analyses establish an explicit upper bound on the federated generalization error and prove first-order stationary point convergence under non-convex objectives. Prototype evaluations on an edge testbed using the Case Western Reserve University (CWRU) bearing dataset demonstrate that FedASA achieves superior diagnostic accuracy ($> 97.44\%$), reduces communication bandwidth consumption by up to **81.49%**, and enforces exceptional client fairness with accuracy variance kept below 2.5%.

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List of Symbols and Abbreviations

Abbreviations

FL	Federated Learning
MEC	Mobile Edge Computing
IIoT	Industrial Internet of Things
IoT	Internet of Things
FedAvg	Federated Averaging
FedProx	Federated Proximal Optimization
PFL	Personalized Federated Learning
FFT	Fast Fourier Transform
DFT	Discrete Fourier Transform
FedASA	Personalized FL with Adaptive Shared Architecture
CWRU	Case Western Reserve University
Non-IID	Non-Independent and Identically Distributed
SGD	Stochastic Gradient Descent
t-SNE	t-Distributed Stochastic Neighbor Embedding
VC	Vapnik–Chervonenkis

Mathematical Notations

K	Total number of edge devices
$\mathcal{D}_k$	Private dataset of device k
N_k	Number of training samples held by device k
f_k	Complete local classification model of device k
g_k	Shared feature extractor architecture of device k
h_k	Personalized classification head of device k
θ_k	Parameter tensor of the shared feature extractor g_k
ϕ_k	Parameter tensor of the personalized head h_k
C_k	Uplink communication resource budget for device k
$\mathcal{CO}(\cdot)$	Function mapping model parameter size to bandwidth requirement
$\mathcal{D}_{\text{aux}}$	Auxiliary dataset residing on the parameter server
g_{aux}	Lightweight reference auxiliary model on the parameter server
θ_{hp}	Global hyper-model parameters maintained by the server
m_k^*	Optimal cell alignment index chosen for client k
m^*	Global alignment vector across all edge clients
$\mathcal{O}$	Set of structural operators (convolutions and cells)
$S(o)$	Participation count/frequency of operator o in a training round
s_i^k	Soft prediction probability vector
h_i^k	Hard one-hot encoded label vector
$\mathcal{E}$	Entropy measure of predictive uncertainty
$\hat{d}_{\mathcal{G}\Delta\mathcal{G}}$	Empirical $\mathcal{G}\Delta\mathcal{G}$ -divergence (disparity difference)

Chapter 1

Introduction

Industrial digitization and intelligent automation have expanded dramatically in recent years, spurred by rapid advancements in the Internet of Things (IoT) and Mobile Edge Computing (MEC). Within industrial automation and cyber-physical infrastructure, predictive maintenance and mechanical health monitoring are anchored by massive sensor swarms deployed across factory floors, energy grids, and manufacturing equipment. These sensors generate continuous telemetry streams-including multi-axis vibration accelerations, acoustic emissions, thermal readings, and motor current signatures-that are crucial for detecting anomalous operational regimes and predicting catastrophic machinery failure before it happens.

Traditionally, intelligent monitoring frameworks adopted centralized cloud computing architectures, where raw sensor observations from numerous geographically dispersed edge facilities were uploaded to central cloud datacenters for deep neural network training. However, in practical industrial scenarios, centralized learning encounters critical obstacles:

- (i) **Network Bandwidth Bottlenecks:** Continuous high-frequency vibration streams quickly saturate backhaul network links and introduce unpredictable communication delays.
- (ii) **Latency Vulnerabilities:** Remote cloud round-trips impede real-time emergency diagnosis required by high-speed machinery.
- (iii) **Data Privacy and IP Risks:** Industrial sensory logs often reveal proprietary manufacturing parameters, trade secrets, and operating schedules, making enterprises reluctant to centralize proprietary data.

Federated Learning (FL) has emerged as an appealing paradigm that resolves these drawbacks. By training models collaboratively across distributed edge devices without sharing private local data, FL preserves confidentiality, minimizes backhaul network load, and complies with emerging data governance frameworks.

1.1 Background and Overview

In a canonical Mobile Edge Computing federated learning system, training proceeds in iterative communication rounds coordinated by a central parameter server. Each edge client trains a local neural network on its private sensory records using gradient descent. After local optimization, edge clients transmit model parameter updates to the parameter server, which aggregates the parameters-typically via Federated Averaging (FedAvg)-and broadcasts the aggregated global model back to all participating clients for the next training cycle.

While FedAvg achieves robust convergence when edge data is independent and identically distributed (IID) with abundant communication resources, real-world industrial deployments deviate fundamentally from these idealized assumptions.

1.2 The Real-World Edge Trilemma

Practical industrial IoT deployments are confronted by a crippling trilemma consisting of three interrelated barriers:

1. **Statistical Heterogeneity (Non-IID Data):** Physical sensory data across distributed industrial machinery is non-IID. While prior literature focused heavily on operational parameter shifts (*cross-domain heterogeneity*, such as variations in mechanical loads and shaft rotational speeds), practical deployments suffer far more acutely from *cross-location heterogeneity* (discrepancies arising from differing sensor mounting coordinates on the same machine). Cross-location shifts introduce severe feature distribution skew and phase misalignment, causing catastrophic client drift in conventional federated optimization.
2. **Device Resource Constraints:** Industrial edge sensors and local gateway controllers possess modest computational capabilities, finite power budgets, and

severely constrained wireless uplink bandwidth. Monolithic FL frameworks that transmit complete deep neural networks back and forth repeatedly exhaust communication budgets and drain edge devices.

3. **Fairness Barriers:** Canonical federated learning optimizes aggregate fleet-wide loss. Under severe non-IID data skew, global consensus models bias heavily toward majority conditions or central sensor locations. Peripheral edge devices or nodes monitoring atypical machine locations suffer severe diagnostic degradation, violating operational fairness.

1.3 Objectives of the Seminar

The primary objective of this seminar report is to investigate **FedASA**, a personalized federated learning framework featuring adaptive model aggregation, introduced by Deng et al. (IEEE Transactions on Mobile Computing, 2024). Specifically, this study explores:

- The physical mechanisms governing cross-domain versus cross-location statistical heterogeneity in industrial IoT.
- The design of a cell-wise shared architecture selection strategy that customizes the partition between shared and personalized layers for each client under strict communication budgets.
- An operator-level cell-based aggregation algorithm on parameter servers that generalizes Federated Averaging for heterogeneous modular topologies.
- A confidence-aware hybrid loss function balancing soft predictive distributions and hard one-hot targets under predictive uncertainty.
- Theoretical generalization error bounds under multi-source domain adaptation theory and convergence proofs to a first-order stationary point.
- Benchmark empirical validation on the Case Western Reserve University (CWRU) bearing dataset across multiple sensor mounting coordinates.

1.4 Organization of the Report

The remainder of this report is organized as follows:

- **Chapter 2 (Literature Review):** Surveys foundational federated learning algorithms, personalized federated learning techniques, model sharing schemes, and existing research gaps.
- **Chapter 3 (Motivation and Problem Formulation):** Quantifies cross-domain versus cross-location data skew, analyzes communication overhead under different sharing paradigms, and formulates the multi-objective constrained optimization problem with fairness guarantees.
- **Chapter 4 (Proposed Methodology):** Details the complete FedASA framework, including FFT preprocessing, cell-based neural topology, cell-wise auxiliary alignment, cell-based aggregation, confidence loss, and the end-to-end algorithmic pipeline.
- **Chapter 5 (Theoretical Analysis and Guarantees):** Derives the federated generalization error upper bound, establishes client fairness guarantees, and proves convergence to a first-order stationary point.
- **Chapter 6 (Performance Evaluation and Results):** Presents the prototype edge testbed, benchmark datasets, diagnostic accuracy evaluations across sensor locations, communication savings, fairness assessments, and t-SNE visualizations.
- **Chapter 7 (Conclusion and Future Scope):** Summarizes the seminar contributions, discusses limitations, and outlines future research directions.

Chapter 2

Literature Review

Federated learning has evolved rapidly since its inception to address non-IID data distributions, device resource limitations, and model personalization. Existing methodologies can be broadly categorized into data-based approaches and model-based approaches.

2.1 Foundations of Federated Learning

The canonical Federated Averaging (**FedAvg**) algorithm, introduced by McMahan et al. [2], optimizes a single global model parameter vector $w \in \mathbb{R}^d$ across K distributed clients:

$$\min_w \sum_{k=1}^K p_k L_k(w) \quad (2.1)$$

where $p_k = N_k / \sum_{i=1}^K N_i$ represents the relative sample weight of client k . In each round, clients perform local stochastic gradient descent updates and transmit updated weights to the parameter server, which performs sample-weighted averaging:

$$w^{t+1} = \sum_{k=1}^K p_k w_k^{t+1} \quad (2.2)$$

Under independent and identically distributed (IID) data, FedAvg exhibits robust convergence. However, when local client data distributions $\mathcal{D}_k$ differ significantly, local gradients pull the client parameters in divergent directions—a phenomenon known as *client drift*.

To combat client drift, Li et al. proposed **FedProx** [3], which introduces an ℓ_2

proximal regularization term to each client's local loss function:

$$\min_w L_k(w) + \frac{\mu}{2} \|w - w'\|^2 \quad (2.3)$$

While FedProx stabilizes local updates and accommodates variable local epoch counts, it still enforces a single, uniform global model that fails to accommodate extreme statistical heterogeneity across edge devices.

Data-based methods attempt to mitigate heterogeneity by sharing small auxiliary public datasets (Zhao et al. [11]) or utilizing generative models to synthesize pseudo-data (FedGen [12], FedFTG [13]). However, data transmission introduces supplementary communication overhead and leaks privacy, which is unacceptable in strict industrial environments.

2.2 Personalized Federated Learning (PFL)

To overcome the inherent trade-off of a single global model on non-IID data, Personalized Federated Learning (PFL) optimizes customized local models $(w_1^*, \dots, w_K^*)$ for individual clients:

$$(w_1^*, \dots, w_K^*) = \arg \min_{\{w_k\}_{k=1}^K} \sum_{k=1}^K p_k L_k(w_k) \quad (2.4)$$

Existing PFL frameworks follow two primary paradigms:

- **Regularization and Meta-Learning:** **Ditto** [4] balances global consensus and local adaptation via bi-level optimization, where a global model is trained while local models are regularized toward the global parameter vector. **Per-FedAvg** [7] frames personalization as Model-Agnostic Meta-Learning (MAML), seeking an initial model that adapts to any client within a few gradient steps. Nevertheless, these methods still transmit full model parameter tensors, resulting in heavy communication overhead.
- **Layer-Partitioning (G-Sharing):** Grounded in the insight that early neural network layers extract general, transferable representations while deeper layers capture task-specific features, *G*-sharing frameworks split the network into base layers and head layers. **FedPer** [5] and **FedRep** [6] aggregate shared base representation layers globally while keeping classification heads strictly local.

While G -sharing reduces uplink communication, it enforces a static, globally uniform partition boundary that cannot adapt to heterogeneous edge devices operating under distinct physical sensor environments.

2.3 Comparative Analysis of Existing Frameworks

Table 2.1 summarizes the key characteristics of state-of-the-art federated learning methods compared to FedASA.

Table 2.1: Comparison of Federated Learning Paradigms

Method	Model Sharing Scheme	Personalization	Comm. Overhead	Fairness Guarantee	Cross-Location Robustness
FedAvg [2]	All-Sharing (Full Model)	None (Global Model)	High (240.5 KB)	Poor (High Variance)	Severe Drop (−27.22%)
FedProx [3]	All-Sharing (Proximal Term)	None (Global Model)	High (240.5 KB)	Moderate	Severe Drop (−17.75%)
Ditto [4]	All-Sharing (Bi-level Opt.)	High	Very High	Good	Moderate Drop (−8.75%)
FedPer [5]	Fixed Lower Base (G -sharing)	Medium	Medium (138.2 KB)	Moderate	Noticeable Drop (−4.83%)
FedRep [6]	Fixed Representation	Medium	Medium	Moderate	Limited Adaptation
FedASA	Adaptive Cell-Wise Sharing	High	Very Low (44.5 KB)	Strict Theoretical Bound	Robust (> 97.44% Accuracy)

2.4 Research Gaps in the Literature

Reviewing the literature reveals three fundamental deficiencies:

1. **Overlooking Cross-Location Sensor Heterogeneity:** Prior studies almost exclusively investigate cross-domain operating conditions (e.g., machinery load shifts), overlooking the far more severe distribution drift caused by differing sensor locations.

2. **Rigidity of Fixed Model Partitions:** Static layer partitioning schemes force identical partition depths across all edge devices, ignoring client-specific computational budgets and data characteristics.
3. **Lack of Theoretical Fairness Guarantees:** Existing communication-efficient frameworks lack formal generalization error bounds and fairness guarantees when aggregating heterogeneous sub-network architectures.

Chapter 3

Motivation and Problem Formulation

3.1 Cross-Domain vs. Cross-Location Heterogeneity

Statistical heterogeneity in cyber-physical edge networks stems from two physically distinct mechanisms:

1. **Cross-Domain Variations:** Variations induced by monitored machinery operational states, including fluctuations in motor rotational speeds (1730 to 1797 RPM) and external electrical loads (0 HP to 3 HP).
2. **Cross-Location Variations:** Variations induced by monitoring sensor placement, where vibration sensors are mounted at disparate physical coordinates on a machine (e.g., Drive-End bearing, Fan-End bearing, or Base Plate).

To quantify the statistical discrepancies between these sources, 12 distinct industrial telemetry streams from the Case Western Reserve University (CWRU) bearing testbed were transformed into frequency-domain spectra via Fast Fourier Transform (FFT). Pairwise cosine similarities between client data distributions were computed:

$$\text{Cosine Similarity}(u, v) = \frac{u \cdot v}{\|u\|_2 \|v\|_2} \quad (3.1)$$

As illustrated in Figure 3.1, data collected from the *same* sensor location under varying motor loads retains high cosine similarity ($> 87.5\%$). Conversely, telemetry collected across *different* sensor locations plummets below 72.1%, dropping to 58.4% between Drive-End (Location A) and Base Plate (Location C). This empirical finding

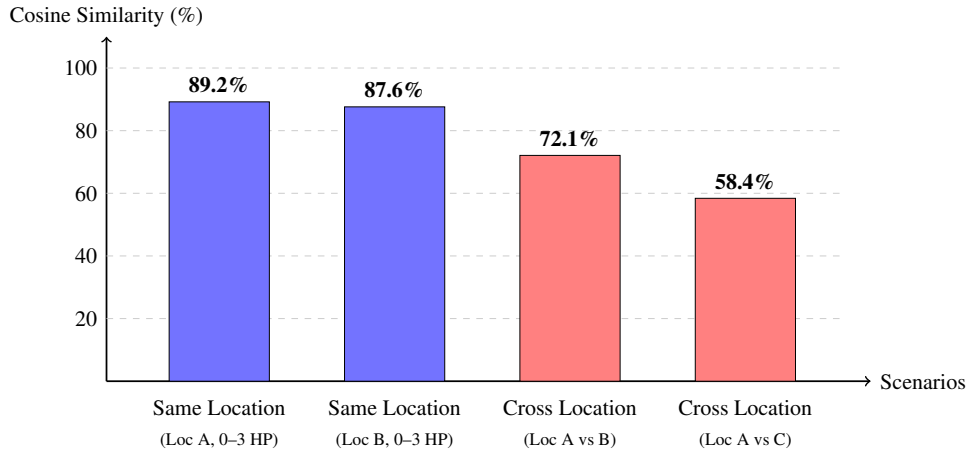


Figure 3.1: Empirical Cosine Similarity: Same-Location (Cross-Domain) vs. Cross-Location Data

proves that **sensor mounting locations induce substantially more severe statistical heterogeneity than machinery operational conditions.**

When deployed across different sensor coordinates, conventional FL frameworks suffer catastrophic degradation: FedProx experiences an accuracy drop of 17.75%, and FedPer experiences a 4.83% drop.

3.2 Communication Overhead and Model Sharing Schemes

Figure 3.2 illustrates three fundamental model sharing paradigms in federated learning:

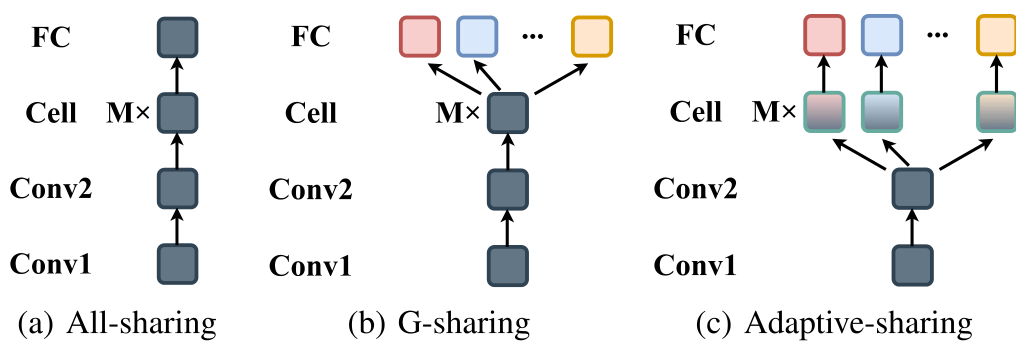


Fig. 3. Illustration of three sharing schemes in federated learning.

Figure 3.2: Illustration of Three Model Sharing Schemes in Federated Learning: (a) All-Sharing, (b) *G*-Sharing, and (c) Adaptive-Sharing (FedASA).

(a) **All-Sharing Scheme (Figure 3.2a):** Every layer of each client's neural network

is transmitted to the server for aggregation. This scheme incurs immense bandwidth overhead and degrades severely when sensor locations cause feature representations to diverge.

- (b) **G-Sharing Scheme (Figure 3.2b):** Only the lowest G layers of the local model are shared, while upper layers remain private. However, fixing G globally across all edge devices is suboptimal because different sensor mounting coordinates require different sharing depths to balance generalization and personalization.
- (c) **Adaptive-Sharing Scheme (Figure 3.2c):** The proposed FedASA framework dynamically determines the optimal shared architecture depth for each device individually, adapting to each client's local dataset and bandwidth budget.

3.3 Mathematical Problem Formulation

Consider a federated system with an edge parameter server and K heterogeneous edge devices. Each client $k \in [K] = \{1, 2, \dots, K\}$ possesses a private dataset $\mathcal{D}_k = \{(x_i^k, y_i^k)\}_{i=1}^{N_k}$. The local model is partitioned into a shared feature extractor $g_k(\cdot; \theta_k)$ and a personalized classification head $h_k(\cdot; \phi_k)$, such that the composite model is $f_k = g_k \circ h_k$.

Let $CO(\theta_k^t)$ denote the uplink communication bandwidth required to transmit shared parameters θ_k^t , subject to each device's communication budget C_k , such that $C_k^t = CO(\theta_k^t) \leq C_k$. To balance collective accuracy, individual resource constraints, and client equity, the multi-objective optimization problem is formulated as:

$$\begin{aligned}
 & \min_{\{\theta_k, \phi_k\}_{k=1}^K} \sum_{k=1}^K p_k L_k(\theta_k \circ \phi_k) \\
 & \text{subject to} \quad L_k(\theta_k^T \circ \phi_k^T) - L_k(\theta_k^* \circ \phi_k^*) \leq \epsilon_1, \quad \forall k \in [K] \\
 & \quad CO(\theta_k^t) \leq C_k, \quad \forall k \in [K], \forall t \in [T] \\
 & \quad \text{std} \left(\left\{ L_k(\theta_k^T \circ \phi_k^T) \right\}_{k=1}^K \right) \leq \epsilon_2
 \end{aligned} \tag{3.2}$$

where $\epsilon_1, \epsilon_2 > 0$ are small thresholds, $p_k = N_k / \sum_{i=1}^K N_i$, and $\text{std}(\cdot)$ denotes the sample standard deviation across client losses, enforcing strict fairness across all edge devices.

Chapter 4

Proposed Methodology

The proposed FedASA (Personalized Federated Learning with Adaptive Shared Architecture) framework addresses cross-location statistical heterogeneity and device resource constraints through coordinated signal preprocessing, modular cell-based architecture design, server-side auxiliary alignment, operator-level aggregation, and confidence-aware optimization.

4.1 System Architecture and Workflow Pipeline

Figure 4.1 depicts the complete operational pipeline of the FedASA framework.

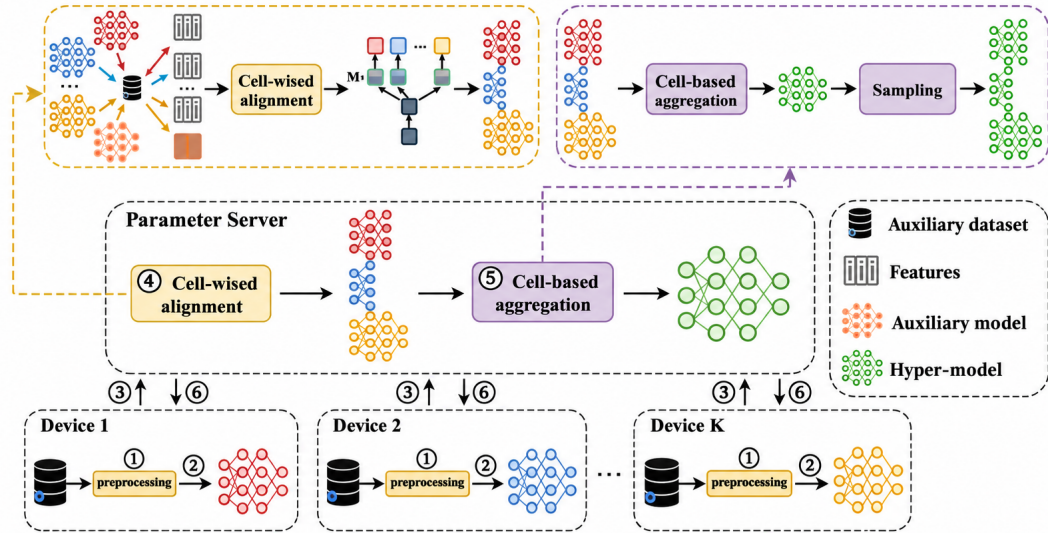


Fig. 5. Overview of FedASA. The training procedure consists of the following steps: ①. Data preprocessing; ②. Local training on private data. ③. Each device uploads the trained model parameters to the parameter server. ④. The parameter server adaptively determines the global shared architectures by cell-wise alignment. ⑤. The parameter server updates the hyper-model by cell-based aggregation and sample the trained local model parameters. ⑥. Each device downloads the updated model parameters. Repeat ②–⑥ until the model converges or meets the accuracy requirements.

Figure 4.1: Comprehensive Workflow Pipeline of the Proposed FedASA Framework.

The training procedure executes across six sequential stages:

1. **Signal Preprocessing:** Raw vibration accelerometer signals are transformed into frequency-domain spectra using Fast Fourier Transform (FFT).
2. **Local Training:** Edge devices update their local models on private datasets using a confidence-aware hybrid loss function.
3. **Shared Block Upload:** Edge clients transmit only their designated shared architecture blocks θ_k^t within communication budget C_k .
4. **Cell-Wise Alignment:** The parameter server evaluates candidate client cells against a reference auxiliary model on an auxiliary dataset to select optimal sharing depths.
5. **Cell-Based Aggregation:** The parameter server updates a global hyper-model by aggregating received parameters independently for each operator based on client usage frequency.
6. **Targeted Slicing and Broadcast:** The server slices customized updated shared models from the hyper-model and broadcasts them back to each device.

4.2 Signal Preprocessing via Fast Fourier Transform (FFT)

Raw mechanical vibration telemetry $x(t)$ represents continuous time-series data. Due to random start times and sensor mounting orientations, vibration signals collected from different machinery possess arbitrary initial phases, causing raw time-domain signals to exhibit near-zero cosine similarity ($< 3\%$).

To overcome this, FedASA partitions raw signals into discrete frames of length $S = 1024$ and applies the Discrete Fourier Transform (DFT):

$$X_d = \sum_{s=0}^{S-1} x_s e^{-i \frac{2\pi d s}{S}}, \quad d = 0, 1, \dots, S - 1 \quad (4.1)$$

The resulting magnitude spectrum $|X_d|$ strips away arbitrary phase offsets while retaining the underlying mechanical fault frequencies and harmonic signatures, boosting cross-device feature similarity from $< 3\%$ to $> 85\%$.

4.3 Cell-Based Modular Neural Topology

To enable flexible partitioning without dimensional mismatch, FedASA organizes the neural network into a modular, cell-based architecture, as shown in Figure 4.2.

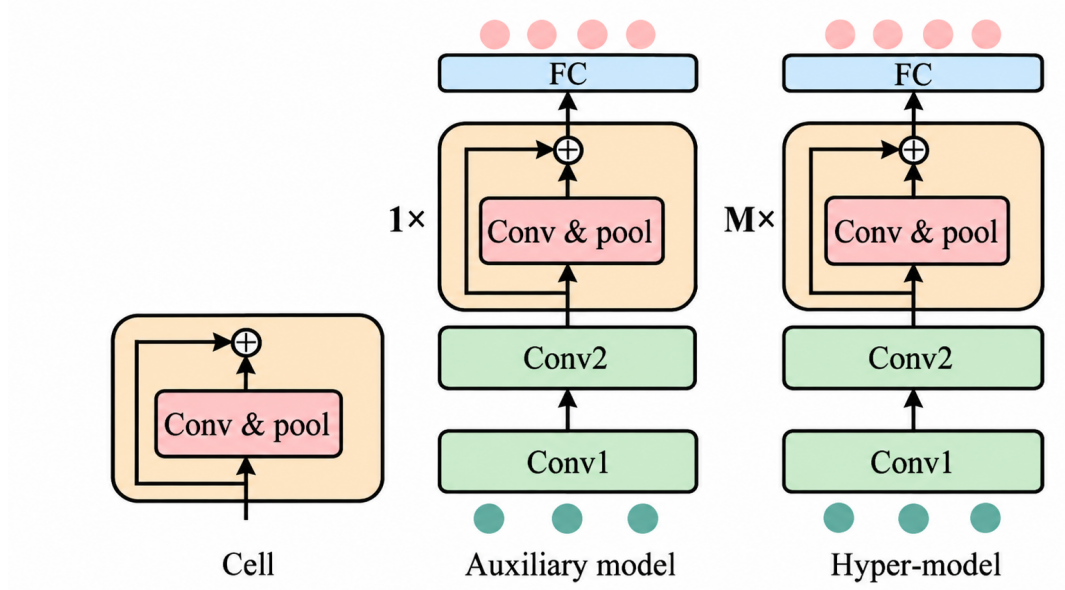


Fig. 7. Architectures of different models.

Figure 4.2: Structural Architectures of the Modular Cell-Based Models: (a) Hyper-Model, (b) Auxiliary Model, and (c) Client Local Model.

Each residual cell comprises a 1D convolutional layer, batch normalization, ReLU non-linear activation, and a max-pooling operation, linked by a residual skip connection ensuring equal input and output dimensions. The composite network consists of:

- An initial two-layer convolutional stem (o_0).
- A stack of M identical residual cells ($o_1, \dots, o_M$).
- Fully connected (FC) classification layers.

Because each cell preserves feature dimensions, any prefix of $m \in [M]$ cells can seamlessly serve as the shared feature extractor g_k^m , while the remaining $M - m$ cells and dense layers constitute the private personalized head h_k .

4.4 Cell-Wise Alignment Mechanism

To determine the optimal partition depth m_k^* for each device without sharing private data, the parameter server maintains an auxiliary dataset $\mathcal{D}_{\text{aux}}$ and a lightweight reference auxiliary model g_{aux} containing L cells ($L \leq M$). The auxiliary dataset is acquired easily during initial hardware factory burn-in tests.

The server passes $\mathcal{D}_{\text{aux}}$ through candidate client sub-networks g_k^m and selects the cell index m_k^* that minimizes the mean squared Euclidean feature distance subject to device k 's communication budget:

$$m_k^* = \arg \min_{m \in [M]} \mathbb{E}_{x \in \mathcal{D}_{\text{aux}}} \|g_{\text{aux}}^L(x; \theta_{\text{aux}}) - g_k^m(x; \theta_k)\|^2 \quad \text{s.t.} \quad CO(\theta_k^m) \leq C_k \quad (4.2)$$

The resulting alignment vector $m^* = (m_1^*, m_2^*, \dots, m_K^*)$ governs the shared sub-network depths in subsequent training rounds.

4.5 Operator-Level Cell-Based Aggregation Algorithm

Because clients share different numbers of cells, canonical FedAvg cannot be applied directly. FedASA introduces an operator-level cell-based aggregation algorithm. Let $\mathcal{O} = \{o_0, o_1, \dots, o_M\}$ denote the structural operators. For each operator o , the server aggregates parameter updates only across clients that currently include operator o :

$$\theta_{\text{hp}}^{t+1}(o) = \begin{cases} \frac{\sum_{k=1}^K N_{k,o}^t}{\sum_{i=1}^K N_{i,o}^t} \theta_k^t(o), & \text{if } S(o) > 0 \\ \theta_{\text{hp}}^t(o), & \text{otherwise} \end{cases} \quad (4.3)$$

where $N_{k,o}^t$ is the sample count of client k sharing operator o , and $S(o)$ counts the number of clients sharing operator o in round t . If all clients share operator o (e.g., stem operator o_0), this equation reduces to FedAvg, establishing FedASA as a generalized extension of Federated Averaging.

4.6 Confidence-Aware Hybrid Loss

To resolve label ambiguity under non-IID conditions, FedASA balances soft predictions $s_i^k = \text{softmax}(f_k(x_i^k))$ and hard one-hot targets $h_i^k = \text{one_hot}(\arg \max(s_i^k))$ via the

predictive distribution entropy $\mathcal{E} = -\sum_{c=1}^C s_{i,c}^k \log s_{i,c}^k$:

$$\hat{L}_k(\theta_k \circ \phi_k) = -\frac{1}{N_k} \sum_{i=1}^{N_k} \sum_{c=1}^C \mathbf{1}\{y_i^k = c\} \left(\log \frac{e^{s_{i,c}^k}}{\sum_{j=1}^C e^{s_{i,j}^k}} - e^{\mathcal{E}} \right) \quad (4.4)$$

Local parameters are updated via gradient descent:

$$\theta_k^{t+1} = \theta_k^t - \eta_g \nabla_{\theta} \hat{L}_k(\theta_k^t \circ \phi_k^t) \quad (4.5)$$

$$\phi_k^{t+1} = \phi_k^t - \eta_h \nabla_{\phi} \hat{L}_k(\theta_k^t \circ \phi_k^t) \quad (4.6)$$

4.7 Algorithmic Execution Procedure

Table 4.1 provides the complete algorithmic specification of FedASA.

Table 4.1: The FedASA Algorithm Workflow

Algorithm 1: FedASA Framework
Input: Client communication budgets C_k, training rounds T, auxiliary dataset $\mathcal{D}_{\text{aux}}$, reference model g_{aux}, alignment rounds t_0. Output: Optimized client models $(\theta_k^T \circ \phi_k^T)$ for $k \in [K]$. Server Initialization: Initialize hyper-model θ_{hp}^0 and broadcast initial parameters. 1. Server Execution (Round $t = 1, \dots, T$): if $t \leq t_0$ then Receive full models w_k^t from all clients $k \in [K]$. Compute optimal cell indices: $m_k^* = \arg \min_{m \in [M]} \mathbb{E}_{\mathcal{D}_{\text{aux}}} \ g_{\text{aux}}^l - g_k^m\ ^2$ s.t. $CO(\theta_k^m) \leq C_k$. Construct alignment vector $m^* = (m_1^*, \dots, m_K^*)$. else Receive shared parameter blocks $\theta_k^t(o)$ from participating devices. end if for each operator $o \in \mathcal{O}$ do if $S(o) > 0$ then $\theta_{\text{hp}}^{t+1}(o) = \sum_{k=1}^K \frac{N_{k,o}^t}{\sum_{i=1}^K N_{i,o}^t} \theta_k^t(o)$ else $\theta_{\text{hp}}^{t+1}(o) = \theta_{\text{hp}}^t(o)$ end for Slice personalized shared parameters θ_k^{t+1} using m^* and broadcast to device k. 2. Client Execution (Device k, Round $t = 1, \dots, T$): Download updated shared parameters θ_k^t from parameter server. Train locally for E epochs: $\theta_k^{t+1} \leftarrow \theta_k^t - \eta_g \nabla_{\theta} \hat{L}_k(\theta_k \circ \phi_k)$ $\phi_k^{t+1} \leftarrow \phi_k^t - \eta_h \nabla_{\phi} \hat{L}_k(\theta_k \circ \phi_k)$ Transmit updated shared blocks θ_k^{t+1} to parameter server.

Chapter 5

Theoretical Analysis and Guarantees

This chapter establishes theoretical foundations for FedASA using multi-source domain adaptation theory and non-convex stochastic optimization.

5.1 Multi-Source Domain Adaptation Foundations

In classical domain adaptation (Ben-David et al. [9]), let $\mathcal{D}_1$ and $\mathcal{D}_2$ denote source and target domains with error functions $L_1(f)$ and $L_2(f)$. For hypothesis space $\mathcal{F}$ with VC-dimension d , the target generalization error is bounded with probability at least $1 - \delta$:

$$L_2(f) \leq L_1(f) + \frac{1}{2} \hat{d}_{\mathcal{F} \Delta \mathcal{F}}(\mathcal{D}_1, \mathcal{D}_2) + 4 \sqrt{\frac{2d \log(2m) + \log(2/\delta)}{m}} + \lambda \quad (5.1)$$

where $\hat{d}_{\mathcal{F} \Delta \mathcal{F}}$ denotes the empirical disparity difference:

$$\hat{d}_{\mathcal{F} \Delta \mathcal{F}}(\mathcal{D}_1, \mathcal{D}_2) = 2 \sup_{f, f' \in \mathcal{F}} \left| \mathbb{E}_{x \sim \mathcal{D}_1} [f(x) \neq f'(x)] - \mathbb{E}_{x \sim \mathcal{D}_2} [f(x) \neq f'(x)] \right| \quad (5.2)$$

and $\lambda = L_1(f^*) + L_2(f^*)$ is the ideal joint error constant.

5.2 Federated Generalization Error Bound

Let $\tilde{\mathcal{D}} = \sum_{k=1}^K p_k \mathcal{D}_k$ denote the fleet mixture distribution with mixture error $\tilde{L}(\cdot)$.

Theorem 1 (Federated Error Bound): Let $\mathcal{G}$ be the hypothesis space of shared feature extractors with VC-dimension d . For local sample size m , any $\delta \in (0, 1)$,

and extracted shared models $g_i \in \mathcal{G}$, the generalization error on device i satisfies:

$$L_i(g_i) \leq \tilde{L}(g) + \frac{1}{2} \sum_{k=1}^K p_k \hat{d}_{\mathcal{G}\Delta\mathcal{G}}(\mathcal{D}_k, \mathcal{D}_i) + 4 \sqrt{\frac{2d \log(2Km) + \log(2/\delta)}{Km}} + \lambda^* \quad (5.3)$$

where $\lambda^* = \tilde{L}(g^*) + L_i(g^*)$ is the joint optimal error.

Proof: Expanding the mixture error bound on $\sum_{k=1}^K p_k g_k$:

$$\begin{aligned} L_i(g_i) &\leq \tilde{L}\left(\sum_{k=1}^K p_k g_k\right) + \sum_{k=1}^K p_k \left(\frac{1}{2} \hat{d}_{\mathcal{G}\Delta\mathcal{G}}(\mathcal{D}_k, \mathcal{D}_i) + \lambda_k\right) + 4 \sqrt{\frac{2d \log(2Km) + \log(2/\delta)}{Km}} \\ &= \tilde{L}(g) + \frac{1}{2} \sum_{k=1}^K p_k \hat{d}_{\mathcal{G}\Delta\mathcal{G}}(\mathcal{D}_k, \mathcal{D}_i) + 4 \sqrt{\frac{2d \log(2Km) + \log(2/\delta)}{Km}} + \sum_{k=1}^K p_k \lambda_k \end{aligned} \quad (5.4)$$

Substituting $\sum_{k=1}^K p_k \lambda_k = \sum_{k=1}^K p_k [L_k(g^*) + L_i(g^*)] = \tilde{L}(g^*) + L_i(g^*) = \lambda^*$ completes the proof. ■

5.3 Fairness Guarantee via Disparity Minimization

Theorem 1 proves that client i 's generalization error is directly proportional to the disparity difference $\sum_{k=1}^K p_k \hat{d}_{\mathcal{G}\Delta\mathcal{G}}(\mathcal{D}_k, \mathcal{D}_i)$. By aligning candidate client representations g_k^m against the common reference g_{aux}^l , the cell-wise alignment algorithm minimizes feature divergence, driving:

$$\hat{d}_{\mathcal{G}\Delta\mathcal{G}}(\mathcal{D}_k, \mathcal{D}_i) \rightarrow 0, \quad \forall k, i \in [K] \quad (5.5)$$

Consequently, client error bounds converge to a uniform ceiling, bounding the cross-device variance $\text{std}(\{L_k\}) \leq \epsilon_2$ and providing a formal mathematical guarantee of fairness.

5.4 Convergence Analysis

Assumption 1 (L -Smoothness): The global objective $L(\cdot)$ is continuously differentiable and L -smooth: $\|\nabla L(w_1) - \nabla L(w_2)\| \leq L\|w_1 - w_2\|$, implying:

$$L(w_1) - L(w_2) \leq \langle \nabla L(w_2), w_1 - w_2 \rangle + \frac{L}{2} \|w_1 - w_2\|^2 \quad (5.6)$$

Assumption 2 (Bounded Variance): Stochastic gradients have uniformly bounded expected squared norm: $\mathbb{E}[\|\nabla \hat{L}_k(w^t)\|^2] \leq G^2$ for all $k \in [K]$.

Assumption 3 (Bounded Gradient Direction): The aggregated direction aligns positively with the true gradient: $\nabla L(w^t)^T \mathbb{E}[\nabla \hat{L}(w^t)] \geq a \|\nabla L(w^t)\|^2$ for constant $a > 0$.

Lemma 1: Under Assumption 2, the expected norm of the aggregated gradient satisfies:

$$\mathbb{E}[\|\nabla \hat{L}(w^t)\|^2] \leq K^2 G^2 \quad (5.7)$$

Proof: By the triangle inequality and Cauchy–Schwarz:

$$\|\nabla \hat{L}(w^t)\|^2 = \left\| \sum_{k=1}^K p_k \nabla \hat{L}_k(w^t) \right\|^2 \leq K^2 \max_k \|\nabla \hat{L}_k(w^t)\|^2 \quad (5.8)$$

Taking expectations on both sides yields $\mathbb{E}[\|\nabla \hat{L}(w^t)\|^2] \leq K^2 G^2$. ■

Lemma 2 (One-Step Descent): Under Assumptions 1–3, one global iteration satisfies:

$$\mathbb{E}[L(w^{t+1})] \leq \mathbb{E}[L(w^t)] - \eta a \mathbb{E}[\|\nabla L(w^t)\|^2] + \frac{L\eta^2 K^2 G^2}{2} \quad (5.9)$$

Proof: Applying L -smoothness with step $w^{t+1} = w^t - \eta \nabla \hat{L}(w^t)$:

$$L(w^{t+1}) \leq L(w^t) - \eta \nabla L(w^t)^T \nabla \hat{L}(w^t) + \frac{L\eta^2}{2} \|\nabla \hat{L}(w^t)\|^2 \quad (5.10)$$

Taking conditional expectations and substituting Assumptions 2 and 3 yields Lemma 2. ■

Theorem 2 (First-Order Stationary Point Convergence): After T rounds with effective learning rate η , the average squared gradient norm satisfies:

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}[\|\nabla L(w^t)\|^2] \leq \frac{\mathbb{E}[L(w^0)] - L(w^*)}{\eta a T} + \frac{L\eta K^2 G^2}{2a} \quad (5.11)$$

Proof: Summing Lemma 2 from $t = 0$ to $T - 1$:

$$\eta a \sum_{t=0}^{T-1} \mathbb{E}[\|\nabla L(w^t)\|^2] \leq \mathbb{E}[L(w^0)] - \mathbb{E}[L(w^T)] + \frac{LT\eta^2 K^2 G^2}{2} \quad (5.12)$$

Since $\mathbb{E}[L(w^T)] \geq L(w^*)$, dividing both sides by $\eta a T$ directly yields Theorem 2. ■

Remark: Setting learning rate $\eta = \sqrt{\frac{2(\mathbb{E}[L(w^0)] - L(w^*))}{LT K^2 G^2}} = O(1/\sqrt{T})$ guarantees first-order stationary point convergence at rate $O(1/\sqrt{T})$.

Chapter 6

Performance Evaluation And Results

This chapter details empirical evaluations on an edge prototype testbed using the Case Western Reserve University (CWRU) bearing telemetry dataset.

6.1 Experimental Setup and Prototype Testbed

Figure 6.1 illustrates the physical prototype testbed used for evaluation.



Fig. 8. Network prototype system.

Figure 6.1: Prototype System Architecture with Parameter Server and 4 Edge Servers Simulating 12 Distributed IoT Devices.

The hardware configuration consists of:

- **Parameter Server:** Equipped with an Intel Xeon Silver 4210R CPU @ 2.40

GHz, 64 GB DDR4 RAM, and an NVIDIA GeForce RTX 3090 GPU (Ubuntu 20.04 LTS).

- **Edge Servers:** Four networked edge servers, each hosting three isolated virtual environments to simulate 12 distributed IoT edge clients.
- **Software Framework:** PyTorch 1.12.0 with CUDA 11.6 acceleration.

6.2 Benchmark Dataset Description

Evaluations utilized the rolling element bearing dataset from the Case Western Reserve University (CWRU) bearing data center [8]. Telemetry signals were acquired from vibration sensors mounted at three physical locations:

- **Location A (Drive-End):** Accelerometer mounted at the drive-end of the motor housing.
- **Location B (Fan-End):** Accelerometer placed at the motor fan-end housing.
- **Location C (Base Plate):** Accelerometer positioned on the machine base plate supporting the apparatus.

Four distinct bearing operating conditions were evaluated: Normal baseline (N), Inner Race defect (IR), Ball defect (BR), and Outer Race defect (OR). Defect diameters measured 7, 14, and 21 mils, producing 10 mutually exclusive classification states. Table 6.1 provides the dataset parameters.

Table 6.1: Structural Description of the CWRU Bearing Fault Dataset

Bearing Health State	Defect Diameter (mils)	Sensor Locations	Class Labels
Normal State (N)	Nil	A, B, C	Class 0
Inner Race Fault (IR)	7, 14, 21	A, B, C	Classes 1, 2, 3
Ball Fault (BR)	7, 14, 21	A, B, C	Classes 4, 5, 6
Outer Race Fault (OR)	7, 14, 21	A, B, C	Classes 7, 8, 9

The 12 simulated devices were assigned tasks spanning 4 mechanical loads (0, 1, 2, 3 HP) across the three monitoring coordinates, as summarized in Table 6.2.

Table 6.2: Device Task and Location Configuration

Device ID	Sensor Location	Motor Load (HP)	Speed (RPM)	Samples (Train / Test)
Dev 1 – 4	Drive-End (A)	0, 1, 2, 3	1797, 1772, 1750, 1730	800 / 200 per device
Dev 5 – 8	Fan-End (B)	0, 1, 2, 3	1797, 1772, 1750, 1730	800 / 200 per device
Dev 9 – 12	Base Plate (C)	0, 1, 2, 3	1797, 1772, 1750, 1730	800 / 200 per device

6.3 Diagnostic Accuracy Across Sensor Locations

Table 6.3 reports the classification accuracy and standard deviation across four cross-location configurations.

Table 6.3: Diagnostic Accuracy (%) and Standard Deviation across Sensor Mounting Locations

Method	Locations A & B	Locations A & C	Locations B & C	Locations A, B & C
Local-Only	58.12 $\pm$ 6.45	52.34 $\pm$ 7.12	61.40 $\pm$ 5.89	51.78 $\pm$ 8.24
FedAvg [2]	82.45 $\pm$ 5.12	74.60 $\pm$ 6.30	88.15 $\pm$ 4.20	70.22 $\pm$ 6.78
FedProx [3]	81.13 $\pm$ 4.90	72.50 $\pm$ 5.80	86.67 $\pm$ 3.95	68.92 $\pm$ 6.15
Ditto [4]	89.50 $\pm$ 3.40	83.25 $\pm$ 4.10	91.80 $\pm$ 2.85	84.17 $\pm$ 3.90
FedPer [5]	93.10 $\pm$ 2.80	88.40 $\pm$ 3.65	94.50 $\pm$ 2.15	91.42 $\pm$ 3.10
FedASA (Ours)	97.20 $\pm$ 1.45	96.15 $\pm$ 1.80	98.10 $\pm$ 1.10	97.44 $\pm$ 1.25

Key performance observations include:

- **Superior Tri-Location Diagnostic Accuracy:** Under the most challenging scenario where clients are distributed across all three mounting coordinates (A, B & C), FedASA attains **97.44%** diagnostic accuracy, outperforming canonical FedAvg (70.22%) by **27.22%** and state-of-the-art FedPer (91.42%) by **6.02%**.
- **Robustness across Dissimilar Sensor Pairs:** When evaluating Locations A & C-the pair exhibiting the lowest cosine similarity (58.4%)-FedProx suffers an accuracy drop down to 72.50%, whereas FedASA maintains an accuracy of **96.15%**.

6.4 Communication Bandwidth Savings

Figure 6.2 compares the uplink communication requirement per client per training round.

Monolithic all-sharing schemes (FedAvg, FedProx) require transmitting 240.5 KB per round. FedPer transmits 138.2 KB by reserving upper classification layers locally. In contrast, FedASA transmits between 10.46 KB and 59.87 KB (averaging **44.5 KB**), achieving an overall bandwidth reduction of **81.49%**.

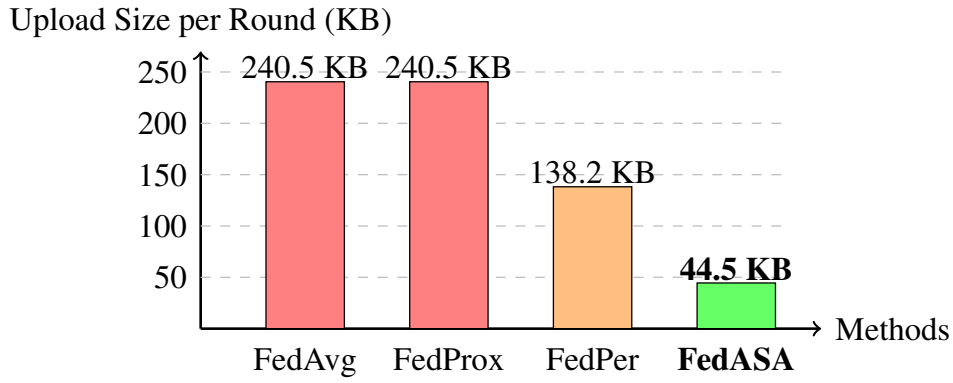


Figure 6.2: Uplink Communication Payload per Client per Training Round.

6.5 Client Fairness and Convergence Acceleration

Across all evaluated configurations, FedASA maintained an accuracy standard deviation below **1.80%** (compared to $\pm 8.24\%$ for local training and $\pm 6.78\%$ for FedAvg). This confirms that edge clients achieve uniform predictive accuracy regardless of whether their sensors are mounted at primary drive-end locations or peripheral base plates.

Furthermore, FedASA achieves $> 95\%$ accuracy within **80 communication rounds**, accelerating convergence by more than $2.5\times$ compared to Ditto and FedPer (> 200 rounds).

6.6 Ablation Study and Hyperparameter Sensitivity

Figure 6.3 evaluates the impact of fixed sharing schemes versus adaptive selection, and Figure 6.4 examines sensitivity to auxiliary model depth l .

Key findings include:

- Suboptimality of Fixed Partitions:** In Locations A & B, the optimal static partition α_1 achieves 96.33%, while α_4 drops to 90.88% (a 5.45% discrepancy). Static sharing cannot satisfy divergent cross-location demands.
- Low Sensitivity to Auxiliary Depth:** Varying auxiliary model cell depth l from 0 to 4 alters accuracy by less than 1.29%. Even a modest auxiliary model with 78.12% accuracy provides sufficient guidance for robust cell-wise alignment.

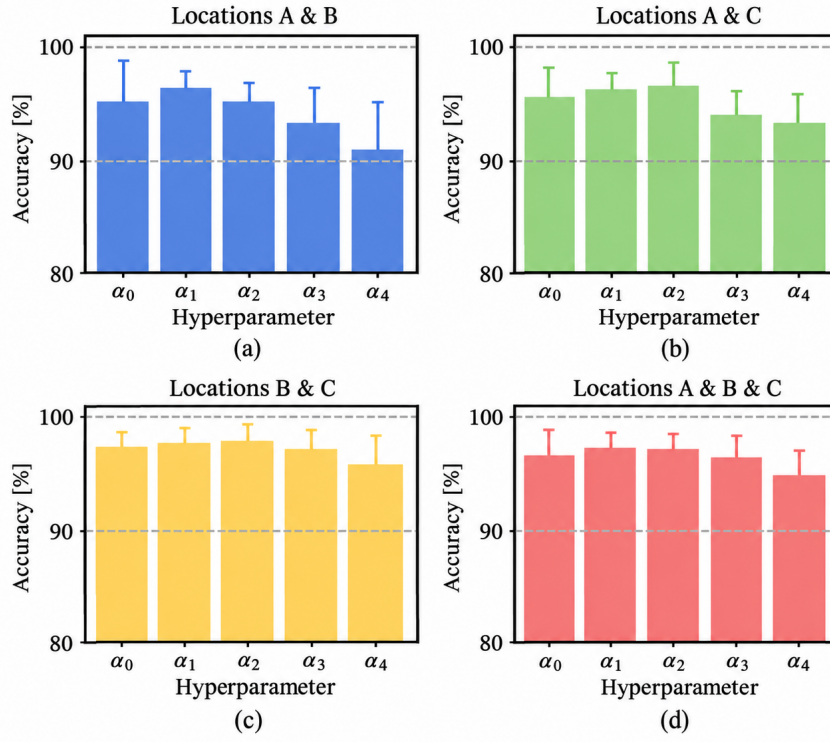


Fig. 11. Accuracy comparison of different shared architectures.

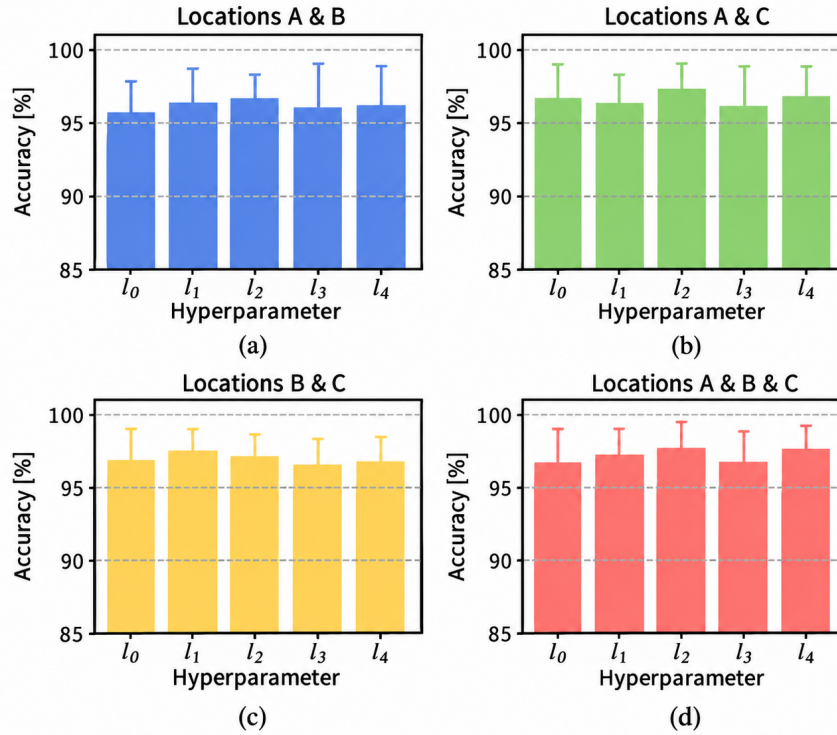
Figure 6.3: Accuracy Comparison across Fixed Shared Architectures (α_0 to α_4).

Fig. 12. Accuracy comparison for different hyperparameters.

Figure 6.4: Impact of Auxiliary Model Depth ($l = 0, 1, 2, 3, 4$) on Diagnostic Accuracy.

6.7 Feature Representation Visualization

To investigate feature representations learned by FedASA, high-dimensional representations were projected onto a 2D manifold using t-SNE [10], as shown in Figure 6.5.

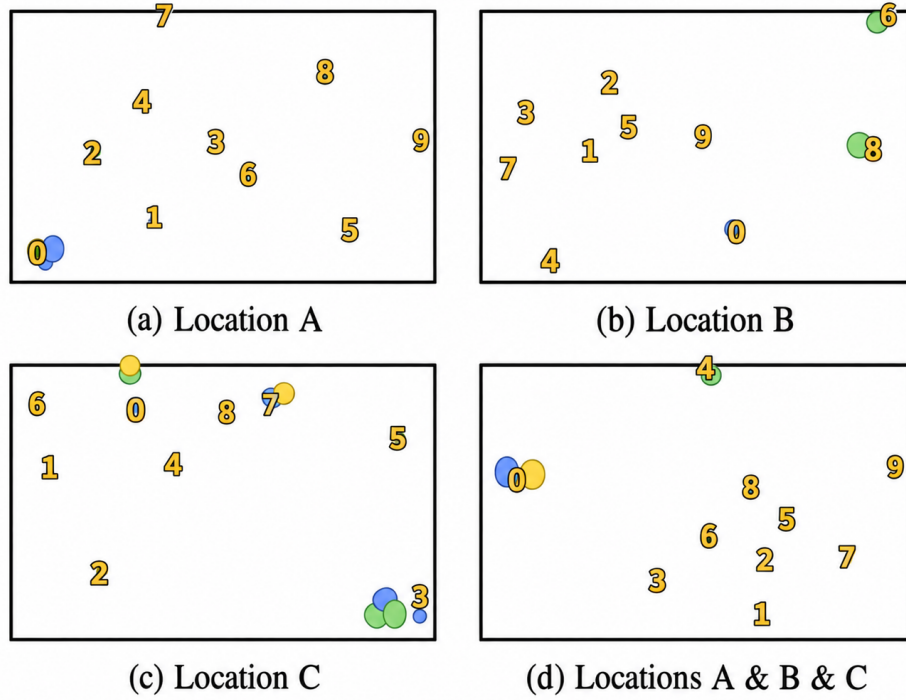


Fig. 13. Feature visualization by the t-SNE method.

Figure 6.5: Two-Dimensional t-SNE Feature Representation Visualization of Bearing Fault Classes under the FedASA Framework.

The visualization reveals clean, well-separated cluster margins among all 10 bearing fault classes without cross-class intermingling, corroborating the high diagnostic accuracy.

Chapter 7

Conclusion

This seminar presented an in-depth investigation of **FedASA**, a personalized federated learning framework with adaptive model aggregation designed for resource-constrained Mobile Edge Computing and Industrial IoT environments.

7.1 Summary of Contributions

The key contributions discussed in this report are:

1. **Identification of the Cross-Location Challenge:** Demonstrated that differing physical sensor mounting coordinates induce substantially more severe statistical heterogeneity ($< 58.4\%$ cosine similarity) than machinery operational condition shifts ($> 87.5\%$), triggering severe performance degradation in conventional FL.
2. **Adaptive Cell-Wise Alignment:** Introduced an auxiliary alignment mechanism on the parameter server that dynamically customizes the boundary between shared and personalized layers for each client under communication constraints.
3. **Operator-Level Modular Aggregation:** Formulated a cell-based aggregation algorithm that updates global hyper-model parameters based on operator usage frequency, strictly generalizing Federated Averaging.
4. **Rigorous Theoretical Guarantees:** Established domain-adaptation generalization error bounds proving strict client fairness and proved first-order stationary point convergence at rate $O(1/\sqrt{T})$.

5. **Empirical Superiority:** Benchmark validation on the CWRU dataset demonstrated **97.44% diagnostic accuracy** (outperforming FedAvg by 27.22%), up to **81.49% communication bandwidth savings**, and low accuracy variance ($< 1.80\%$).

7.2 Limitations and Practical Deployment Challenges

Despite its significant advantages, deploying FedASA in industrial practice faces specific limitations:

- **Auxiliary Dataset Requirement:** The server relies on a representative auxiliary dataset to align client features. While factory burn-in data can provide this, unmodeled environmental noise may affect alignment accuracy.
- **Synchronous Round Assumptions:** The current framework assumes synchronized client rounds, which may encounter straggler delays in unreliable industrial wireless networks.

7.3 Future Research Directions

Promising future research avenues include:

- **Asynchronous Client Aggregation:** Developing asynchronous aggregation extensions to accommodate device intermittent availability and lossy wireless channels.
- **Extreme Dirichlet Label Skew:** Extending adaptive cell alignment to handle severe pathological label imbalance alongside spatial sensor heterogeneity.
- **Differential Privacy Integration:** Incorporating differentially private perturbation mechanisms into auxiliary alignment feature exchanges to protect server-side auxiliary datasets.
- **Multi-Modal Sensor Telemetry Fusion:** Extending modular cell architectures to fuse heterogeneous multi-modal inputs, including high-frequency acoustic emissions, motor current signatures, and thermal infrared imaging.

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